

PRODUCT & CAPABILITY DATASHEET

Data Alchemist

Agentic Data Quality & Trust Platform — capability, connector, and governance reference

Document type: Product & Capability Datasheet

Prepared for: SCALEX Pursuit 2026 — Supporting Artefact

Coverage: 11 modules · 5 database connectors · 5 LLM providers · governance model

Module-by-Module Capability Reference

Module	What it does	AI-assisted?
Profiling	Autonomous data quality profiling — 11-check LangGraph pipeline: nulls, duplicates (whole-row and key-based), referential orphans, schema drift, format patterns, numeric distributions	Narrative only — checks are deterministic
Rule Studio	DQ rule authoring — AI-recommended rules and natural-language-to-SQL conversion, both gated by 7 static-analysis checks and human approval	Yes
DQ Execution	Runs approved rules as deterministic SQL, computes weighted per-table quality scores, captures sample failed records	No — by design
Anomaly Inbox	Volume, distribution, segment, and freshness anomaly detection with plain-English root-cause narratives	Yes
Trust Dashboard	Cross-module trust score, CDE health, and an AI usage/cost transparency panel	Aggregation only
Impact Graph (Lineage)	Table- and column-level lineage discovery (declared FK + query-log inference), downstream-impact narratives	Yes
Scenario Simulator	Sandboxed "what-if" incident injection, LLM classification of scenario type, zero writes to production tables	Yes
Pre-run Advisory / Trust Receipt	Predicted trust score before a pipeline run; a per-query data-quality "nutrition label"	Yes
Dictionary & CDEs (Metadata)	AI-suggested column business descriptions with steward approve/edit/reject workflow; Critical Data Element promotion	Yes
Connections	Multi-platform connection management, Fernet-encrypted credentials, live connectivity testing	No
Task Board	Human task queue linked back to the specific rule, anomaly, or CDE that generated it	No

"AI-assisted" reflects whether an LLM call is in the module's normal path — every module's *execution* and *scoring* logic is deterministic regardless.

Connector & LLM Provider Matrix

Database Connectors

Platform	Auth Types	Status
SQL Server	SQL auth, Windows auth, Azure AD	Supported
Snowflake	Username/password, key pair, OAuth	Supported
Databricks	Personal access token, OAuth	Supported
PostgreSQL	Username/password	Supported
DuckDB	File path or in-memory	Supported

Adding a new platform means implementing the 5-method BaseConnector interface and registering it in `registry.py` — no changes anywhere else in the agent, API, or UI layers.

LLM Providers (via LiteLLM gateway)

Provider	Example model	Switch method
Anthropic	Claude Sonnet (default)	LLM_PROVIDER=anthropic
OpenAI	GPT-4o	LLM_PROVIDER=openai
Azure OpenAI	Azure-hosted GPT models	LLM_PROVIDER=azure
Google	Gemini	LLM_PROVIDER=gemini
Ollama (self-hosted)	Any local open model	LLM_PROVIDER=ollama

One environment variable swaps the model platform-wide — no code changes, no redeployment of individual modules. This matters for clients with data-sovereignty or model-choice requirements.

AI Governance & Security Overview

AI Governance

- **7+ explicit human decision gates** across the platform — rule approve/reject, risk suppress/note, metadata dictionary decide, anomaly acknowledge, lineage edge approve/reject, task management.
- **Full cost transparency** — every LLM call is logged with model, token counts, and cost to `ai_usage_log`, visible on the Trust Dashboard's Cost & Governance panel.
- **Zero fine-tuning** — all reasoning runs over live schema and profiling data pulled at request time; no training data pipeline to secure or govern.
- **Structural validation before human review** — LLM-proposed rules pass 7 deterministic static-analysis checks (inverted null checks, uncorrelated EXISTS, unverified table/column references, and more) before a human ever sees them.
- **Deterministic execution path** — rule execution and quality scoring are 100% SQL, with no LLM in the runtime hot path.

Security & Compliance

- **Credential protection:** Fernet symmetric encryption at rest; credentials are never returned by any API response or written to logs.
- **Authentication:** bcrypt password hashing and JWT bearer tokens for local accounts; Microsoft Entra ID SSO (OAuth2 Authorization Code + PKCE) for enterprise identity.
- **Tenant isolation:** 28 of 29 satellite database tables carry a `connection_id` foreign key — every module operates within that scope by construction.
- **Auditability:** an immutable `audit_trail` table records every human decision with before/after value snapshots.
- **Data minimization:** connectors read schema metadata and bounded samples for profiling; there is no bulk-export path off the platform.